

SuperDoc math: m:box / m:borderBox spec coverage test

Each scenario below tests a specific branch of ECMA-376 §22.1.2.11-14 and §22.9.2.7. Labels describe the expected MathML.

A. m:borderBox defaults

1. No borderBoxPr (expect notation=box)

abc

2. Empty borderBoxPr (expect notation=box)

$abc2$

B. ST_OnOff value variants (on m:hideTop)

3. m:val=1 (expect top hidden)

$T1$

4. m:val=true (expect top hidden)

Ttr

5. m:val=on (Annex L.6.1.3 uses this — expect top hidden)

Ton

6. Bare tag hideTop/ (expect top hidden — spec default val=1)

Tbr

7. m:val=0 (expect NOT hidden)

$T0$

8. m:val=false (expect NOT hidden)

Tf

C. Individual hide flags in isolation

9. hideBot only (expect top left right)

B

10. hideLeft only (expect top bottom right)



11. hideRight only (expect top bottom left)



12. hideTop + hideRight — spec §22.1.2.12 example (expect bottom left)



13. All four hides, no strikes (expect fallback to mrow, no border)

$mrow$

D. Strike directions (all borders hidden)

14. strikeBLTR alone (BL to TR = slash = updiagonalstrike)

$BLTR$

15. strikeTLBR alone (TL to BR = backslash = downdiagonalstrike)

$TLBR$

16. strikeH alone (horizontalstrike)

H

17. strikeV alone (verticalstrike)

V

E. Strike combinations

18. strikeBLTR + strikeTLBR = X pattern

X

19. All four strikes

$\cancel{\cancel{H}}$

20. Default box + strikeH (expect: box horizontalstrike)

$\boxed{H}$

21. hideTop + strikeH (expect: bottom left right horizontalstrike)

$\boxed{pH}$

F. Spec Annex L.6.1.3 example

22. a squared = b squared + c squared crossed out with TL-BR diagonal, using m:val=on from spec

$a^2 = b^2 + c^2$

G. m:box scenarios

23. Plain m:box with content == (expect mrow)

$a = b$

24. m:box with boxPr/opEmu — spec §22.1.2.13 example (opEmu currently dropped)

$a = b$

25. m:box with boxPr/noBreak (noBreak currently dropped)

$a = b$

26. m:box with all boxPr flags (all currently dropped)

$a = b$

27. Empty m:box (expect null/dropped)

H. Nested structures

28. m:box inside m:borderBox

$a = b$

29. m:borderBox inside m:box

$a = b$

30. Nested m:borderBox inside m:borderBox (outer partial-hide + inner strike)

1az startfirstmiddlesecondend

Commented [TA1]: First inner comment - covers "first"

Commented [TA2]: Second inner comment - covers "second"

Commented [TA3]: Outer comment - should cover entire text